

# QUINLAN A. CAMPBELL

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## EDUCATION

Virginia Tech | Bachelor of Science, May 2027

**Major:** Environmental Data Science | **Minor:** GIS | **Overall GPA:** 3.75

**Relevant Coursework:** Programming in Python, Intro to GIS, Modeling in GIS, Environmental Data Science, Systems Analysis, Remote Sensing, Forest Photogrammetry and Spatial Data Processing, Hydroinformatics, Biological Statistics

## RESEARCH EXPERIENCE

***AI, Drones, and Disaster Impact Research Assistant*** | Geography Department, Virginia Tech (May 2026-Present)

- Analyze drone and aerial imagery to assess road damage and passability following hurricanes.
- Assist in AI model validation and refinement by manually labeling image datasets.

***Hydrochemistry Student Researcher*** | FREC Department, Virginia Tech (Aug. 2025-Present)

- Perform automated watershed delineation in R with LiDAR data and hydrology tools (terra, sf).
- Conduct terrain and slope analyses to classify strike- and dip-slope catchments and assess runoff.
- Support field sampling operations, including water chemistry collection and in situ sensor calibration.

***Water Quality Lab Technician*** | Carey Lab, Virginia Tech (Mar. 2025-Present)

- Collect water quality samples and field observations across reservoir, inflow, and outflow sites.
- Support laboratory operations through field preparation, acid washing, data entry, and chemistry runs.
- Analyze water quality, macrophyte, and ecological monitoring data in R and ArcGIS.

***Co-Lead Researcher- Iliamna corei*** | Restoration Ecology Lab, Virginia Tech (Aug. 2024-Present)

- Conduct fieldwork including site visits, debris clearing, and habitat assessments to support restoration.
- Develop a long-term monitoring and restoration protocol to guide future conservation actions.
- Secure grant funding to implement conservation measures.

***Soybean Laboratory Technician*** | Leisner Lab, Virginia Tech (Jan. 2025–May 2025)

- Conducted comprehensive plant care, including watering and managing soybean sample processing.
- Analyzed soybean samples to investigate the impact of elevated CO<sub>2</sub> on nutrient levels.
- Utilized advanced tools to measure transpiration and assess leaf physiology.

***BIRDNET Research Assistant*** | Restoration Ecology Lab, Virginia Tech (Jan. 2025–May 2025)

- Analyzed acoustic recordings to identify avian species presence across restored and unrestored stream sites.

***Fuel Dynamics Field Research Assistant*** | FREC Department, Virginia Tech; Catoctin Mountain & Manassas Battlefield Park (May 2024–Aug. 2024)

- Assisted with field research across 340 plots in Catoctin Mountain Park and Manassas Battlefield Park.
- Conducted surveys on deadwood and trees, capturing DBH, canopy class, and decay class.
- Demonstrated strong plant identification skills while navigating diverse terrains.

## **ADDITIONAL PROFESSIONAL EXPERIENCE**

### ***Disaster Divisions Intern*** | United Way Southwest Virginia (Jan. 2026-Present)

- Maintain and standardize disaster assistance case data, including household size, case status, and funding identifiers, to support recovery operations.
- Develop geospatial visualizations in ArcGIS to analyze spatial and temporal patterns in disaster assistance, including case clustering, funding intensity, and monthly distribution across service counties.

### ***Communications and Volunteer Engagement Intern*** | Virginia Master Naturalist (Aug. 2024-Dec. 2024)

- Collaborated with program leaders and volunteers to create a social media strategy.
- Contributed to marketing that raised awareness, recruited participants, and strengthened partnerships.
- Supported the VMN virtual conference by engaging volunteers and documenting outcomes.

### ***Student Sustainability Intern*** | Virginia Tech Office of Sustainability (May 2024-Dec. 2024)

- Collaborated with various departments (Energy Management, Stormwater Management, Sustainable Dining, Housing & Residence Life) on technical sustainability projects as part of assigned teams.

## **PUBLICATIONS**

**Campbell, Q.**, Sappestein, B., Hernandez, A., Willis, B., Fitzgerald, M., Lipscomb, M., Polk, N., Reid, R. E. B., Edwards, R., Klopff, R., & Reid, J. L. (in review). Population status and trend of *Iliamna corei* (Peter's Mountain mallow; Malvaceae), a federally endangered Appalachian endemic.

Reid, J. L., **Campbell, Q.**, Duerr, N., Fitzgerald, H. M., Judge, P., Polk, N. D., Ripa, G. N., Sapperstein, B., Staengl, E. J., & Vrh, S. (in review). Undammed: Freeing Rivers and Bringing Communities to Life [Book review]. Restoration Ecology.

Carey, C.C., A. Breef-Pilz, D.X. Edwards, **Q. Campbell**, S.E. Kenny, and E. Walters (2026). Secchi depth data and discrete depth profiles of water temperature, dissolved oxygen, conductivity, specific conductance, photosynthetic active radiation, oxidation-reduction potential, and pH for Beaverdam Reservoir, Carvins Cove Reservoir, Falling Creek Reservoir, Gatewood Reservoir, and Spring Hollow Reservoir in southwestern Virginia, USA 2013-2025 ver 14. Environmental Data Initiative. <https://doi.org/10.6073/pasta/b20ba95d3667532e4dd8bede8a00ab52>

Carey, C.C., A. Breef-Pilz, D.X. Edwards, F.M. Iannucci, **Q. Campbell**, S.E. Kenny, and E. Walters (2026). Time series of dissolved methane and carbon dioxide concentrations for Falling Creek Reservoir, Beaverdam Reservoir and Carvins Cove Reservoir in southwestern Virginia, USA during 2015-2025 ver 10. Environmental Data Initiative. <https://doi.org/10.6073/pasta/2b97469a6278df366478218743ab33bc>

Reid, J. L., Velo, M., Aparicio-Vera, S., **Campbell, Q.**, Colberg, E., Coscia, J. T., Fitzgerald, M., Frelich, L., Hyatt, L., & Logan, C. M. (2025). Tree planting choices mediate wildfire damage to tropical forest restoration in eastern Madagascar. Restoration Ecology. <https://doi.org/10.1111/rec.70251>

Reid, J. L., Burt, M. A., **Campbell, Q.**, Coscia, J. T., Huston, J. C., Kubesch, J. O. C., MacDougal, P., Ripa, G. N., Staengl, E. J., & Viani, R. A. G. (2024). Ecological Restoration: Moving Forward Using Lessons Learned [Book review]. Restoration Ecology. <https://doi.org/10.1111/rec.14163>

## **PRESENTATIONS**

**Campbell, Q.**, Sapperstein, B., Reid, J.L., *Population Status and Trend of Iliamna corei* (Peter's Mountain Mallow; *Malvaceae*), a Federally Endangered Appalachian Endemic. Poster presentation, School of Plant and Environmental Science Symposium, Blacksburg, VA, 2026.

**Campbell, Q.**, Sapperstein, B., Reid, J.L., *Population Status and Trend of Iliamna corei* (Peter's Mountain Mallow; *Malvaceae*), a Federally Endangered Appalachian Endemic. Poster presentation, Dennis Dean Undergraduate Research and Creative Scholarship Conference, Blacksburg, VA, 2026.

**Campbell, Q.**, Gannon, J.P., *Hydrochemical Signatures of Scarp and Dip-Slope Drainage in the Valley and Ridge Region*. Poster presentation, Geological Society of America Joint Southeastern, North-Central, and South-Central Section Meeting, Memphis, TN, 2026.

**Campbell, Q.**, Sapperstein, B., Reid, J.L., *Conservation and Management Procedures for Critically Endangered Peter's Mountain Mallow (Iliamna corei)*. Poster presentation, Biological and Ecological Restoration Initiative Symposium, Blacksburg, VA, 2025.

## **GRANTS AND ACADEMIC HONORS**

CNRE Undergraduate Research Fellowship: \$2,500 (2026)

Ut Prosim Internship Support Fund: \$800 (2026)

Global Change Center Undergraduate Research Grant: \$316 (2024)

Dean's List (Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026)

## **LEADERSHIP, ACTIVITIES, SERVICE, VOLUNTEER WORK**

**Service Chair, Philanthropy Chair, Pledge Parent** | Lambda Iota Mu (Co-ed Service Frat) (Feb. 2024-Present)  
Coordinating and leading service, philanthropy, and new-member initiatives.

**Volunteer** | Campus Kitchen (Nov. 2025-Present)

Coordinating the collection of surplus food from campus dining facilities and transporting it to local food banks.

## **PROFESSIONAL MEMBERSHIPS AND AFFILIATIONS**

Ecological Society of America (ESA)

American Geophysical Union (AGU)

Geological Society of America (GSA)

## **CERTIFICATIONS**

FEMA IS-100.c – Introduction to the Incident Command System

FEMA IS-700.b – National Incident Management System (NIMS)

## **SKILLS**

**Data & Analysis:** R, Python, ArcGIS Pro, GitHub, Excel, geospatial analysis, data visualization

**Water Sampling Methods:** Field filtering, Ultraportable Greenhouse Gas Analyzer (UGGA), Conductivity, Temperature, and Depth (CTD) Device, Fluoroprobe, chlorophyll filtering

**Other Field & Laboratory Methods:** acid washing, plant ID, forest inventory, CN analysis, Li-COR machinery